

Mass as the Compressed State of Space; Gravity Field as Stretched State of Space Induced by Mass Compression; Dark Matter as the Quasi-Compressed State of Space; Both Matter and Dark Matter Originate from the Ultra-Compressed Initial State of the Universe

-Section9-Section13 of the Quantized Elastic Cosmic Model, <https://doi.org/10.5281/zenodo.15278220>

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Abstract:

These Sections presents a novel theoretical framework in which mass arises as a compressed state of space, formed by the local densification of the Space Elementary Quanta (SEQ) network. In this model, gravity is not a fundamental force in the traditional sense, but rather the restoring force induced by the stretching of residual space surrounding a mass-induced compression zone. The gravitational field thus emerges as a consequence of the spatial deformation created by the mass-compressed region, and can be understood as a macroscopic manifestation of the elastic response of space itself.

Furthermore, dark matter is interpreted as a quasi-compressed state of space—a weaker or less stable form of compression compared to ordinary mass. This distinction accounts for its weak interaction with electromagnetic fields and its apparent gravitational dominance in cosmic structures.

Crucially, both ordinary mass and dark matter originate from the ultra-compressed initial state of the early universe, where the primordial SEQ network was in a highly compressed configuration. As the universe expanded, localized compression patterns stabilized into mass and dark matter states.

An essential mechanism in this model is the role of the Higgs field as a chiral quantum lock, which stabilizes the spatial compression state by locking the SEQ configuration into a coherent, long-lasting structure. This locking mechanism prevents the immediate relaxation of the compressed space, thereby enabling the emergence of stable mass. The Higgs field thus acts not merely as a source of mass via coupling, but more fundamentally as a topological stabilizer of spatial structure.

This unified perspective provides a new physical picture for understanding the nature of mass, gravity, and dark matter, and suggests a potential pathway toward the integration of general relativity, quantum field theory, and cosmology under a common spatial-structural framework.

9 Gravitational Interaction, General Relativity and Cosmic Evolution Model

9.1 Gravitational interaction is modeled as a translational action described by matrices that alter the equilibrium spacing between SEQ. SEQ are interconnected via spring-like bonds in their ground state(Refer to the basic setting in Section 1.6, page2). Gravity modifies this spacing, creating tension with finite potential energy. This system behaves like a loaded spring: under gravitational fields, oscillation frequencies decrease, reducing local spacetime transformation rates and causing time dilation - matching general relativity's predictions while revealing its mechanism. Mass-generated gravity acts as a spherically divergent translation with inverse-square density, curving flat spacetime topologically to produce general relativistic metric changes.

Within this discrete framework, the singularity paradox of black holes is naturally resolved because the tension between SEQ has a upper limit.

Macroscopically, (1)It explains metric variations and gravitational time dilation predicted by general relativity, while remaining compatible with its continuity assumption, (2)The ground-state energy of SEQ can also give a depiction of the cosmological constant in general relativity. (3)At the cosmic edge, adjacent SEQ lack outward coordination sites, creating an expansion tendency from interior with the initial kinetic energy released during the birth of the universe—a potential mechanism for cosmic expansion.

9.1.1Under this model, a detailed deduction of the process of cosmic expansion.

1	Potential energy is stored in elastic bonds composed of sub-Planck scale components.
2	In this model the energy of SEQ m_i equals the SEQ resonant kinetic energy plus the elastic potential energy assigned to this SEQ from its adjacent elastic bonds manifested as frequency suppression. Chapter 14 will mention that SEQ spin and resonance frequency mutually excite each other and change synchronously; therefore, SEQ spin kinetic energy can also be represented by SEQ resonant kinetic energy.
3	Chapter 10 will discuss that mass formation is mainly due to spin locking the spatial compression state and the key factor is the coupling confinement potential between the network spinor and the SEQ fixed chirality spin.

Stage-Phase	Stage Name	Process	Universe State	Thermodynamic Characteristics
0-Compression	Pre-Big Bang Initial State	The universe's SEQ network is highly compressed, with resonant frequencies close to zero. The initial low-entropy state may be reflected in a few SEQ having particularly high energy, while most have low energy.	High-energy Aggregation State	Low entropy
1-Compression	Compression Potential Energy → Kinetic Energy	Elastic compression potential energy is released and converted into cosmic expansion kinetic energy	Accelerating Expansion	Low entropy, high energy concentration, rapid entropy increase
2-Stretching	Kinetic Energy → Tension Potential Energy	Expansion kinetic energy is converted into tension potential energy	Decelerating Expansion	Increasing entropy

Stage-Phase	Stage Name	Process	Universe State	Thermodynamic Characteristics
3-Stretching	Tension Potential Energy → Kinetic Energy	Tension potential energy is released and converted into contraction kinetic energy	Accelerating Contraction	Entropy continues to increase
4-compression	Kinetic Energy → Compression Potential Energy	Contraction kinetic energy is converted into compression potential energy	Decelerating Contraction	Entropy continues to increase
5-repeated Oscillation → Equilibrium Oscillation	Energy Homogenization → Equilibrium Oscillation	In each cycle, the energy distribution becomes more uniform, with no obvious concentrated states remaining	Approaching Equilibrium State	Entropy approaches maximum, oscillating universe in thermal equilibrium
This process does not collapse back to the initial birth configuration of universe, nor does it reduce entropy—since the entropy increasing trend remains invariant under expansion or contraction, the homogenization of energy distribution is an irreversible process, until entropy reaches its maximum value.				

- $E_{\text{global SEQ network}} = K_{\text{resonant}}(K_{\text{spin}}) + U_{\text{elastic}}$;
- $U_{\text{elastic}} = U_{\text{compress-stretch}} + U_{\text{twistor}}$; (Space network spinor)
- U_{twistor} (Space network spinor) converts into $K_{\text{resonant}}(K_{\text{spin}})$; embodied as space network spinor

Among them, the conversion of twistor potential energy into resonant kinetic energy corresponds to the Kerr metric. However, since the Kerr metric distortion is not completely radial, the conversion of Kerr metric distortion potential energy into resonant kinetic energy simultaneously alters both the resonant frequency and the resonant axis vector (generalized coordinates in the model mentioned in Section 13.4). Further mathematical formalization requires rigorous derivation by professional physicists based on observational data; here, only a rough model can be proposed.

- $U_{\text{compress-stretch}}$ converts into $K_{\text{resonant}}(K_{\text{spin}})$

Supplementary description of the initial low-entropy state of the universe

Before the Big Bang, the universe was in a highly compressed state. The elastic potential energy was divided into two components: one was spatial compression potential energy, and the other was spatial twistor potential energy (Space network spinor). The spatial twistor potential energy was concentrated in some local regions of space — this constituted the initial low-entropy state. After the Big Bang, the twistor potential energy from this tiny region was released through conversion into SEQ resonant kinetic energy and SEQ spin kinetic energy, and this energy was then conducted to other regions of space. This process represents the specific mechanism of entropy increase.

9.1.2 The mass formation and matter-energy conversion mechanisms in different stages of cosmic evolution:

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- First stage of cosmic expansion: mass generation dominates.

According to this quantized elastic space-time model, mass is formed when compressed states are locked by the coupling of SEQ network spinors and the fixed chirality spin of SEQ. The formation of mass states depends on the degree of spatial compression and the intensity of SEQ network spinor. Since the universe as a whole is in a highly compressed phase during this stage, and SEQ network spinor is concentrated on a local zone, it is consistent that mass is formed. In the first stage, the entire cosmic space is generally in a highly compressed state, and during the initial inflationary period, the compression is even more pronounced. Due to the space network spinor-SEQ fixed chirality coupling locking mechanism (similar to the Higgs mechanism, introduced in Chapter 10), the highly concentrated spinor of local zone causes certain compressed spatial states to be locked into mass states. Therefore, mass generation dominates in this stage, especially during the early inflationary phase, when the spatial compression level is higher and better matches the compression required for mass state formation.

- Second and third stages of cosmic evolution:

transformation from mass states to energy states dominates, with possible new mass state formation during the process.

In these stages, the universe as a whole enters a stretched phase. According to this model, mass states become unstable due to the stretching of the space itself. The conditions for mass state become weaker, leading to more mass states breaking down. During this phase, new elements—mass states—may be generated within massive celestial bodies.

- Fourth stage of cosmic evolution : the universe re-enters a compressed phase.

However, at this stage, the distribution of SEQ network spinor distribution has become more uniform, and the effect of the Higgs-like mechanism is weaker. Although new mass states may still form, the overall rate of mass generation is significantly lower than in the first stage.

- Fifth stage of cosmic evolution:

repeated oscillations leading into thermal equilibrium oscillation.

As the network energy becomes increasingly uniformly distributed—an irreversible entropy increase process—the proportion of mass states in the universe gradually decreases through repeated oscillations. Eventually, all mass states are transformed into energy states, meaning the spin kinetic energy distribution of the SEQ network becomes uniform, and the universe enters oscillation in a thermal equilibrium state.

This mechanism, in which high spatial compression collaborates with space network spinor to generate mass, may also explain the observed **anomalies in galaxy rotation curves**. Specifically, the early universe featured a high concentration of network spinor. This intense network spinor concentration, coupled with the extreme spatial compression, facilitated the formation of mass.

Consequently, the resulting mass structures and their surrounding space likely retain higher-than-expected spinor, which could account for the observed rotational dynamics in galaxies.

Dark matter formation is also dominated by a synergistic locking mechanism involving spatial compression states and a Higgs-like mechanism (space network spinor and SEQ fixed-chirality spin coupling), similar to the mass generation mechanism in the early stages of cosmic expansion. However, dark matter formed at a later stage than ordinary mass. The spatial compression level during this phase was insufficient to reach the threshold required for mass generation. Moreover, by this time, the increase in entropy had led to a more uniform distribution of space network spinor, reducing the strength of the spinor field in the Higgs-like mechanism to a level insufficient to lock high-compression states. This is the fundamental difference between dark matter and mass states. In essence, dark matter is also a form of locked spatial compression state, but with a much lower compression strength compared to mass states. Nevertheless, it still exerts a stretching effect on the surrounding space, manifesting as a gravitational field.

Why is there a global gravitational tension-restoring potential in the first stage of cosmic expansion, even though the entire space is in a compressed phase? Because the compression level within mass structures is significantly higher than that of the surrounding space, leading to a large gradient in spatial deformation, which gives rise to a relative tension-restoring potential — the gravitational potential.

This explanation is self-consistent and visually clear, especially in providing a concrete physical image and mechanism for the large-scale generation of mass states in the early inflationary universe.

9.2 Consistent with general relativity, high-velocity or accelerated transformations of localized matter compress space, thereby inducing tensile stretching of surrounding space (equivalent gravitational effects). This process not only induces spacetime curvature but also modulates the conduction frequency of waves.

In relativity theory, equivalent gravitational fields from velocity/acceleration (1) share the same core mechanism as mass-generated gravity—local SEQ compression (reduced spacing) inducing spacetime stretching, but (2) exhibit vector-directional dependence (anisotropic compression) instead of spherical symmetry, and (3) reflect inertial-SEQ lattice coupling through their non-spherical divergence and preferred orientation alignment in quantum spacetime structure.

9.3 Detailed Correspondence with Newton's law of universal gravitation

"The number of SEQ on the surface of a mass source corresponds to the number of gravitational flux lines (i.e., the count of gravitational transmission paths). As the gravitational field diverges spherically, the density of these flux lines becomes inversely proportional to the surface area at any given radius, thus exhibiting an inverse-square relationship with distance. This result directly coincides with Newton's law of universal gravitation."

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9.4 Correspondence with Newton's First Law—the Law of Inertia

In this framework, the compressed space at the front and the stretched space at the rear—both caused by the object's motion at Constant Velocity—are always the interfacial boundary between the matter wave and the surrounding SEQ. Beyond this boundary, there is no compression or stretching induced by the object's motion at Constant Velocity, only a shift in the harmonic vibration phase of the SEQ, the system reaches equilibrium at Constant Velocity.

The work input during acceleration establishes interfacial strain energy in the SEQ network, which then sustains uniform motion through elastic potential equilibration.

9.5 Understanding on General Relativity

9.5.1 Under gravitational and equivalent gravitational interactions, the dynamic deformation of 3D space structural matrix and variation in local SEQ density distribution corresponds to Metric field in General Relativity.

9.5.2 Minimum cumulative conduction count path adjustment along with the cumulative dynamic paths connecting every two-points with the minimal count of adjacent SEQ through spacetime distortion corresponds to geodesic path in general relativity. (Principle of least action)

9.5.3 Global topological homeomorphic transformation in SEQ framework corresponds to Diffeomorphism invariance in General Relativity.

9.5.4 The continuity assumption in general relativity, analogous to the continuum medium framework in fluid mechanics, constitutes a necessary and effective computational framework.

9.5.5 Black hole event horizon:

Inside the event horizon of a black hole, due to intense gravitational forces, the spacing between SEQ is compressed to its limit. This extreme compression approximately and locally halts: (1)Energy conduction (2) space transformations (3)Entropy increase step (Neglecting black hole accretion).

9.5.6 Gravitational and Kinematic Time Dilation

All factors that induce metric variation, including mass (gravity), velocity, and acceleration(equivalence principle), compress or stretch spacetime locally, thereby modulating the transformation frequency of related space. This frequency suppression constitutes the fundamental mechanism of time dilation.

10 Mass, Gravity, SU(3) and Higgs field in Quantum Field Theory

10.1 SU(3) as the Origin of Mass Derivation

10.1.1 General Relativity establishes that gravitational fields manifest as metric perturbations →spacetime curvature.

10.1.2 Mass must therefore induce localized spacetime distortion→creating the observed gravitational potential.

10.1.3 This implies mass itself represents a condensed form of spacetime deformation → self-consistent with stress-energy sourcing curvature.

10.1.4 Within hadrons, quark-gluon dynamics are governed by SU(3) color interactions→the dominant force compressing spatial quanta (SEQ).

10.1.5 Thus, SU(3)-mediated compression of SEQ networks → generates both quark confinement energy (mass) and external spacetime stretching (gravity).

10.1.6 Generalizing this mechanism→ equivalent effects (velocity/acceleration) anisotropically compress local space→inducing equivalent gravitational attraction via adjacent SEQ tension.

10.2 U(1): Electromagnetic Interaction

10.3 SU(2): Adjustments of Rotational Axes , spin and encoding chirality in Charged Microscopic Particle Structural Matrices. Encoding Rotational chirality of Charged Microscopic Particle Structural Matrices as the Origin of Weak Interaction Symmetry Breaking. because the rotational Structural Matrices with different chirality have different coupling mode with the fixed chirality of SEQ's ground-state spin. Although SU(2) lacks a chirality modulation parameter, its definition as adjusting rotation axes and spin for charged particles with specific chirality intrinsically encodes chiral variables.

10.4 SU(3): Note: The phase in SU(3): compression or stretching shift between SEQ.

10.4.1 Imagine the 3D dynamic quasi-spherical matrix structure of quarks as a multi-layered and multi-axial rotational configuration. Due to the high-energy concentration within the structure, the SEQ within the structure remain in a dynamic equilibrium of compression or stretching, while the interactions between layers also maintain a dynamic equilibrium of compression or stretching.

10.4.2 Fractional quark charges emerge from stratified SEQ layers in proton/neutron matrices, with 2/3-charged quarks occupying twice the layers of 1/3-charged quarks. The multi-layered structure well explains the observed differences between high-energy (uniform angular distribution) and medium-energy regimes in electron-proton scattering experiments.

10.4.3 The color property of quarks corresponds to the long axis of their dynamic structural matrix, specifically the axis with the highest energy density distribution within the quark's structural configuration. The color neutrality of protons and neutrons corresponds to the global spatial symmetry, the isotropy of the electric field (protons) and structural stability exhibited by their spatial structural matrices.

10.4.4 Antiquarks correspond to the handedness reverse representation of structural matrix rotational transformation of their corresponding quarks.

10.4.5 The 8 generators of SU(3) correspond to 8 distinct interactions mediated by different gluons. Among them, the 6 non-diagonal matrices represent combinations of color exchange operations, stretching and compression phase transformations with phase variations (3×2); while the 2 diagonal matrices correspond to scaling transformations across the three color dimensions. These gluons and their 8 distinct interaction types operate within the interlayer regions of the multi-layered structural matrices of protons or neutrons.

The dynamic color exchange corresponding to the generators of SU(3) and the three-dimensional color distribution modulation are fundamentally linked to the translational action of gravity. Therefore, the color modulation and exchange interactions of SU(3) could constitute one of the origin of mass.

10.4.6 Gluons mediate compression and tensile stresses between quarks or interlayer SEQ. Gluons can be understood as a kind of quasi-structure of rotational highly condensed SEQ, akin to a high-density array of springs.

10.4.7 Quark asymptotic freedom and color confinement originated from nonlinear variations in compression-tensile tensions among SEQ.

10.4.8 The three-quark point-like configuration inherently fails to achieve spatial symmetry, contradicting the observed spherical charge distribution of protons, whereas this hypothesis of a layered arrangement in a quasi-spherical structure of up and down quarks within the proton offers a more plausible explanation for the integer charge of the proton and the isotropic nature of the

electric field as well.

10.4.9 The discrepancy in the proton's g-factor from theoretical models stems from an underestimation of the gluon field's spinor contribution. If the effect of the rapidly rotating gluon field were properly accounted for, this deviation would significantly diminish. Moreover, the conventional three-point-quark distribution framework fundamentally cannot accommodate a proportionally substantial gluon field spinor component. The layered structure proposed in this model presents a viable architectural framework worthy of serious consideration.

10.5 How SU(3) Generators Mediate Mass Formation

They compress local space while performing 3D modulation, axis transformations, and compression-phase adjustments to ensure the compressed space remains approximately spherically symmetric. As a result, the gravitational field generated by mass (the stretching of external space due to local compression) is also spherically divergent, guaranteeing the isotropy of mass-induced gravity. The physical picture is now clear.

10.6 The essence of mass is the storage of gravitational (spatial elastic) potential energy under the interaction of SU(3) corresponding to the compression of space.

10.6.1 Dimensional analysis dictates that the relationship between mass and energy must satisfy $[E]/[m][v^2]$, with the proportionality coefficient determined by the fundamental constants of spacetime (the speed of light, c).

10.6.2 The compressed potential energy of mass in localized space is inherently mainly released as gravitational waves with radiation, which propagate at speed c , thus directly yielding $\Delta E = \Delta mc^2$.

10.7 Complementary Role of the Higgs Field: Symmetry Breaking and "Locking" Mechanism:

Role of the Higgs Field as quantum chiral lock in mass-gravity generation

What role does the Higgs mechanism play here? If a spring is merely compressed or stretched, it naturally returns to equilibrium. However, the asymmetry of the Higgs mechanism becomes crucial. Anyone familiar with mechanical design would immediately recognize that locking a compressed state requires a latching mechanism — for instance, a torsion spring combined with a ratchet, which stores and locks elastic potential energy. This aligns perfectly with the geometric intuition of the Higgs mechanism's "Mexican hat" potential, much like a rotary knob with a torsion spring or a ratchet that unidirectionally locks compressive energy.

10.7.1 The Higgs field plays a crucial yet subtle role in this framework by acting as a stabilizing "quantum chiral lock" that preserves the compression effects mediated by the SU(3) gluon field on the local SEQ network. While the SU(3) color force actively compresses the local space to generate mass-energy through spatial deformation, the Higgs mechanism serves to maintain this compressed configuration in a stable equilibrium state. This locking function is particularly vital for quark confinement, as it prevents the rapid dissipation of the gluon field's compressive energy that would otherwise lead to deconfinement. The Higgs field's symmetry-breaking properties thus complement the SU(3) compression mechanism by providing an additional interaction of stability to the mass-generating structure. In essence, if the SU(3) mediated compression is likened to a tensed spring storing potential energy, the Higgs field acts as the catch mechanism that keeps the spring compressed, ensuring the persistence of the mass effect. This dual mechanism - active compression by color forces and passive stabilization by the Higgs field - offers a more complete picture of mass generation that bridges quantum chromodynamics with electroweak theory while remaining consistent with the discrete spacetime framework

proposed in the paper. The interplay between these mechanisms may also help explain why certain particles (like quarks) exhibit both confinement and mass properties, while others (like leptons) primarily acquire mass through Higgs interactions alone (like a preloaded torsional spring energy storage combined with a ratchet).

This dual mechanism—where the QCD color interaction-SU(3) acts as a compressive spring system, while the Higgs mechanism functions like a preloaded torsional spring combined with a **ratchet** (enabling unidirectional energy storage while preventing reversal)- provides a vivid mechanical analogy for how fundamental particles maintain their mass stability in the quantum spacetime fabric. Just as a ratchet's teeth enforce unidirectional motion through asymmetric geometry, the Higgs' chiral coupling to the SEQ ground state spin with fixed chirality may similarly lock the gluon field's compressive energy in an metastable configuration.

10.7.2 Origin and Physical Picture of the Higgs Mechanism

In this model, the Higgs mechanism is fundamentally a synergistic effect of gauge fields (U(1), SU(2), SU(3)) rather than an independent field. The torsional-spring-like vorticity of the Higgs mechanism originates from the vorticity coupling of U(1), SU(2), and SU(3), while the ratchet-like locking arises from symmetry breaking induced by the fixed chiral spin of SEQ.

According to the model specification in Section 1.6, the spin degrees of freedom of Space Elementary Quanta (SEQ) and higher-level spinors are decoupled from the elastic bonds between SEQ and their sub-Planckian components. This decoupling mechanism naturally explains the vorticity disparity between the gravitational field (emerging from macroscopic SEQ network stretching) and the gluon field (originating from localized SU(3) compression of SEQs).

10.7.3 Therefore, quark confinement may arise from the combined effects of the Higgs field's quantum chiral lock and the nonlinear response of spatial elasticity(QCD).

10.8 A fundamental duality emerges between the SU(3)-driven compression of matter at quantum scales and the emergent gravitational field: The mass of hadrons arises from intense color-force compaction within subnuclear volumes, whereas gravity manifests as the coherent stretching of the finite space elementary quanta (SEQ) fabric. This stark contrast in interaction ranges — from quark confinement to system-wide SEQ deformation — naturally explains the hierarchical strength difference between nuclear and gravitational forces.

In simple terms, the elastic coefficients and deformation ratios differ between the compressive phase (QCD) and the tensile phase (gravity). The deformation of gravity is distributed across the entire space, whereas QCD's deformation is highly localized. This difference leads to the distinct energy scales of QCD and gravity. Different deformation ratios lead to different strengths.

10.9 In nuclear reactions, the release of kinetic energy primarily corresponds to the elastic potential energy-kinetic energy of the QCD dynamic spring array, while the breaking of the Higgs mechanism mainly releases stored Fermionic Spinor Energy -akin to torsional spring energy storage in the form of radiation. This explains the energy type distribution in nuclear reactions and the radiative phenomena in QED.

A Hypothetical Qualitative Analysis:

In light-nuclei fusion reactions (e.g., D-T fusion), the simple nuclear structure and low mass of light nuclei result in a relatively minor contribution from the "spinor-twisted spacetime structure" (analogous to a torsion-spring energy storage mechanism) induced by Higgs field coupling.

Consequently, the proportion of radiative energy release in total reaction energy remains notably small. In contrast, heavy nuclei (e.g., ^{235}U) possess significantly higher nucleon number density, wherein the Higgs-mediated spinor distortion effects become more dominant. This leads to a markedly increased share of radiative energy release through β -decay chains during fission processes. The observed disparity may reflect enhanced synergy between Higgs field and QCD confinement potential in heavy nuclear structures.

11 Thoughts on the 3D Spatial Arrangement Matrix of Microscopic Particles

11.1 Spatial Arrangement Matrix Representation of Electrons:

To ensure the observed spherical symmetry of the electron's electric field, its structure must comprise at least four or more SEQ in a 3D (possibly multi-layered) configuration. Additionally, the electron's structural matrix may undergo rapid multi-axis rotation. Estimates based on electron mass suggest this matrix contains a large number of SEQ.

11.2 Representation of Electric Charge:

Electric charge may correspond to the intrinsic multi-layered, multi-axis rotational dynamics of the structural matrix governing microscopic particles. All electric charged microscopic particles are embedded with analogous substructures.

11.3 Fractional Charges of Quarks:

Fractional charges cannot exist in isolation but depend on SU(3)-mediated collective effects of quark confinement. The underlying mechanism suggests that when gluons between quarks disintegrate, the quarks must either likewise disintegrate or undergo reintegration.

11.4 Annihilation and Decay of Microscopic Particles:

The annihilation or decay of microscopic particles fundamentally arises from the disintegration or reintegration of their spatial structural arrangement matrices.

11.5 Mechanism Analysis of Positron and other types of Antiparticle Scarcity:

The intrinsic spin of an electron is essentially the orbital rotation of SEQ in the electron's structure around the electron's center.

Positron and other types of Antiparticle scarcity emerges from their interaction with the fixed-chirality ground-state spin of SEQ, inducing instability of their structural matrices, a mechanism that simultaneously explains parity violation.

11.6. Geometric intuition for the half-integer spin of electrons:

It is imaginable that the spin axis of an electron undergoes a flipping motion, with the flipping period synchronized with the spin period. This would allow for a 4π reset, while also providing a geometric intuition for the SU(2) double covering of SO(3). In fact, this combination of spin and axis-flipping corresponds to the mechanism required to explain the spherically symmetric and divergent electric field of the electron.

SU(2) double covers SO(3), meaning that the double covering is a homomorphism where the preimage set contains twice as many elements as the image set. My intuitive understanding of this is in terms of a "double rotational degree of freedom". This double rotational freedom could be interpreted as a combination of spin and axis rotation. How can a 4π reset be realized? When the spin period and the axis-flipping period are synchronized, a reset after two flips of the axis corresponds precisely to a 4π spin reset.

This geometric intuition is consistent with the observed physical phenomena.

11.7 The Nature of Lepton Mass:

In this model, the electron possesses an internal structure, and its spin arises from the field excitation of this structural rotation. Such a spinorial field excitation can distort spacetime, while the near-light-speed tangential velocity of the electron's rotation corresponds to a spatial twist that generates a spherically divergent gravitational field. The mass of the electron is fundamentally the result of the external spacetime stretching—manifested as a gravitational field—induced by this rotating spinorial distortion. This can equivalently be understood as the inertial mass described by relativity.

From this perspective, the electron mass calculated in this theoretical model is inherently inertial mass, and the speed-of-light constraint applies equally to the tangential excitation wave velocity at the electron's surface. Under this understanding and constraint, recalculating the electron's radius yields a self-consistent result where **the tangential velocity never exceeds the speed of light**.

The flaw in traditional theoretical calculations lies in their presupposition that the electron's mass is rest mass, thereby underestimating the relativistic inertial mass. This leads to an overestimated radius and, consequently, a paradoxical derivation of superluminal tangential velocity. In contrast, this model eliminates the paradox: the electron's rest mass approaches zero, is not governed by chromodynamics, and is instead dominated purely by the dynamic spinorial field excitation of the Higgs mechanism—an inertial mass framework free of contradictions.

12 Quantum Gravity, Gravitons and Space Elastic Response Frequency

12.1 Gravity fundamentally stems from its mediation by elastic bonds(sub-Planckian constituents) between SEQ rather than direct SEQ interactions.

12.2 When the resonant frequency of SEQ significantly exceeds the elastic response frequency of inter-SEQ bonds, gravitational field mediation does not encode SEQ's spectral fingerprints.

12.3 The method of gravitational wave frequency detection implies that the detected frequency range should fall within the spatial elastic response frequency range. As our understanding of gravitational wave frequencies expands, so too will our knowledge of the spatial elastic response frequency range.

The sub-Planckian spacetime constituents mediating gravitational interactions may correspond to the gravitons, as they are the fundamental units responsible for conducting spatial elasticity and gravitational effects.

13 Space Deformation - SEQ Resonant Frequency Modulation Duality

-Connecting GR to QFT

In Section 4, we discussed the action and mentioned that spatial deformation corresponds to frequency modulation. In this chapter, we will further explore this topic. Furthermore, the subsequent mathematical formalization of this model does not require a specific characterization of spatial deformation. Instead, it maps spatial deformation onto the corresponding region's frequency domain of SEQ network while distinguishing between compression phases, stretching phases, and twistors. Frequency modulation is used to characterize spatial deformation and potential energy changes.

13.1 Frequency Modulation as an Essential Description of Spatial Deformation

- The model suggests that any metric change in space, such as curvature caused by gravitational fields, modulates the resonant frequency of SEQ. Compression and stretching phases influence frequency domain modulation through asymmetric elastic coefficients. This frequency modulation directly encodes the geometric information of spatial deformation, eliminating the need for additional Riemann geometry descriptions.
- The traditional concept of potential energy terms (gravitational, electromagnetic, or quantum field potentials) is reinterpreted as frequency modulation of SEQ resonance. For instance, a decrease in gravitational potential energy corresponds to a frequency domain offset, while the release of potential energy manifests as dynamic modulation restoring the frequency to its high-frequency ground state. This mapping enables a unified frequency-domain representation of the metric field in general relativity and potential energy terms in quantum field theory.
- Entropy Increase Rate: In addition, since the conduction frequency within a local space directly determines the local entropy increase rate of the system, there also exists a dualistic modulation mechanism between space geometry deformation and the rate of entropy increase. This relationship is self-consistent and analytically derivable under the SEQ quantized space model.

Simplified Pathway for Mathematical Formalization:

13.2 Classification of Spatial Deformations: First, classify spatial deformations into 4 types (specific classifications can be refined based on future research):

- Stretching Phase
- Compression Phase
- Left-handed Twistor
- Right-handed Twistor

Since the model assumes that SEQ have a fixed chirality spin in their ground state, the frequency modulation caused by left-handed and right-handed twistors is not entirely symmetric. Based on QCD and cosmological observations, the elastic coefficients and frequency modulation of space should be nonlinear functions. Therefore, according to existing QCD, electromagnetism, and cosmological observations, preliminary modeling of the operator functions for these 4 deformations to SEQ frequency modulation can be established. Embedding these operators into the action function or other equations allows the analytic expression of action from Chapter 4 to represent the modulation of transmission frequency due to deformation.

13.3 Construction of Discrete Functional Framework:

By defining the local frequency response function of the SEQ network, continuous spatial deformation can be transformed into a parametric problem on discrete frequency lattice points. This model converts the geometrical dynamics of spatial tensor-twistor deformation into frequency dynamics, providing a new mathematical framework for unifying gravity and

quantum theory. Its formal simplicity may offer new tools for physics. Future research should focus on developing specific algorithms for discrete functional equations and establishing mappings with parameters of the Standard Model. However, it should be noted that the specific mathematical modeling of this discrete model depends on the adjacency topology of the space SEQ network, which remains undetermined at present. Therefore, the mathematical phenomenological fitting is only approximate.

13.4 Generalized Coordinates in This Model:

Preliminary Discussion on the Lagrangian and Hamiltonian Analytical Mechanics Framework Based on This Model:

In the topologically homeomorphic space field of this quantized elastic spacetime model, each SEQ (Space Elementary Quantum) has a fixed spatial coordinate and fixed adjacency relations. The generalized coordinates of each SEQ are its resonance frequency and resonance axis vector. These two quantities, combined with the dynamic structural matrix of matter, can generate physical quantities such as mass, momentum, velocity, acceleration, force, kinetic energy, and potential energy—all expressed as polynomials of the generalized coordinates or their differentials (partial derivatives or integrals). Based on this framework, conservation equations can be constructed.

13.5 Under the topologically homeomorphic setting, the spatial coordinates of each SEQ serve as its structural label and constitute important topological invariants. These coordinates remain fixed during dynamical evolution (i.e., they do not change over time), and therefore are not subjected to time differentiation. Nevertheless, they play a crucial role in constructing conservation equations, such as defining local gradients and adjacency relations. Hence, they should be regarded as background structural parameters rather than components of generalized coordinates.

13.6 Mass, Mass represents a spatial compression state that cannot be characterized by a single SEQ. Instead, it requires a local description in terms of the local spatial compression rate — the degree of deformation, which can be quantitatively expressed as the average frequency shift within that local space. Frequency directly reflects the extent of spatial deformation.

Since this model does not incorporate the conventional concept of volume, the notion of compression is inherently tied to the topologically homeomorphic SEQ network. Whether compressed or not, the structure remains a SEQ grid, and thus, spatial compression must be described through either:

A general relativistic tensor representation of the compression rate, or

The average resonance frequency of the SEQs contained within the mass-bearing object, which captures the spatial deformation effect.

Therefore, mass can be effectively represented by the number of SEQs within the object and their collective resonance frequency. This leads to the following formulation:

$$m = K \times N \times (\omega_p / \bar{\omega})$$

Where:

m: mass

K: a dimensional conversion constant (can be dimensionless or carry traditional mass dimensions)

N: number of SEQs contained in the mass-bearing object

$\bar{\omega}$: average resonance frequency (relative to Planck frequency shift)

ω_p : Planck frequency

$(\omega_p / \bar{\omega})$: represents the degree of spatial deformation

13.7 Force — next, we discuss $F = ma$. Acceleration can be understood as the rate of change of spatial deformation gradient between a mass-bearing object and its external environment. This gradient of deformation induces the object to maintain structural synchronization during acceleration, resulting in additional spatial deformation — compression at the front end of acceleration and stretching at the back end. In this framework, m represents the structural response barrier (mass), and F is the deformation action required to induce an additional gradient of spatial deformation in an object of mass m. The essence of $F = ma$ is: in order to cause a change in the spatial deformation gradient (acceleration) of an object that possesses a structural response barrier (mass), a corresponding deformation action (force) must be applied.

13.8 Other conceptual constructions of classical physical quantities — the following are only examples and not strict mathematical formalizations:

Starting from the perspective of spatial deformation — local frequency duality, we define the phase $\phi = \Delta\omega / \omega_0$, where $\Delta\omega$ is the local frequency shift and ω_0 is the reference frequency. From this, we construct the wave vector $k = \nabla\phi$. Then, we define momentum of generalized coordinate-SEQ Frequency as $p = \hbar k$, where $\hbar$ is the reduced Planck constant, $\hbar = h / (2\pi)$.

The resonance frequency and resonance axis vector of each SEQ — which are the generalized coordinates — correspond to two different sets of analytical expressions for generalized momentum. The generalized momentum of the SEQ resonance axis needs to be further developed in future work. According to this model, the direction of the SEQ resonance axis depends on the adjacency configuration, which is discrete and finite, thus making the direction quantized as well.

In the analytical expression of the generalized momentum of the SEQ resonance frequency as a generalized coordinate, the reduced Planck constant $\hbar$ is not used to characterize the quantization of angular momentum, but rather to describe phase quantization in the harmonic oscillation of SEQ (this is only a conceptual analysis tool — the actual phase variation may not necessarily be quantized). This allows for a formal description of the phase gradient between matter and its surrounding SEQ in the process of material wave formation.

Velocity is then defined as $v = p / m$, acceleration as $a = dv / dt$, force as $F = dp / dt$, and so on.

13.9 In this model, the spatial field is a topologically homeomorphic structural field, and the motion of matter corresponds to the propagation of excitation waves on this field. Therefore, when constructing conservation equations, they may differ from the traditional conservation equations in analytical mechanics, but they should be able to derive equivalent forms corresponding to the classical ones. Here, I can only propose a rough framework, and the specific rigorous mathematical formalization likely still requires the work of professional physicists to complete.

Although Einstein adopted the geometric description based on the metric tensor in his General Relativity, he had already noticed the significant influence of gravity on light frequency shifts. The 'Deformation-Frequency Equivalence Principle' proposed in this paper can be regarded as a continuation of this unification vision—it reinterprets the variation of spacetime tensor structure as a frequency modulation process of SEQ, thereby establishing a new unified descriptive paradigm between the microscopic and macroscopic scales.

Deriving a metric-quantum frequency function from cosmological gravitational redshift measurements and QCD-level interactions could establish a functional mapping-

$(\omega_{\text{QFT}} = f(g_{\mu\nu})_{\text{GR}})$ between GR's spacetime geometry and QFT's quantum observables, offering a mathematically tractable and empirically testable tool for unifying gravity and quantum theory.